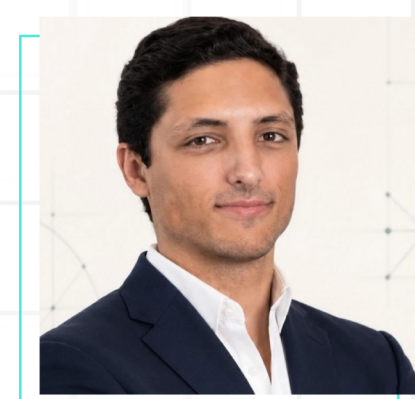


From Design to Digital Twin: Physics and AI for Fatigue-Aware Floating Offshore Wind Towers

From Knowledge to Competitiveness

Portugal-US Partnerships 2026 Summit

MIT Portugal Program



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THE CHALLENGE

- 25-year fatigue life to prove
 - Thousands of simulations per design
 - Loads unmeasured once installed
- Key for Portugal's deep Atlantic waters [1].

OUR APPROACH

The same structural physics governs design, simulation, learning and monitoring.

One family of physics + AI tools on the IEA 22 MW floating reference turbine.

1 DESIGN FLOAT [2]

Redesign the tower for a 25-year fatigue life.

3 SIMULATE LESS NeuroFLOAT [4]

Simulate a physics-selected subset, let AI fill the rest.

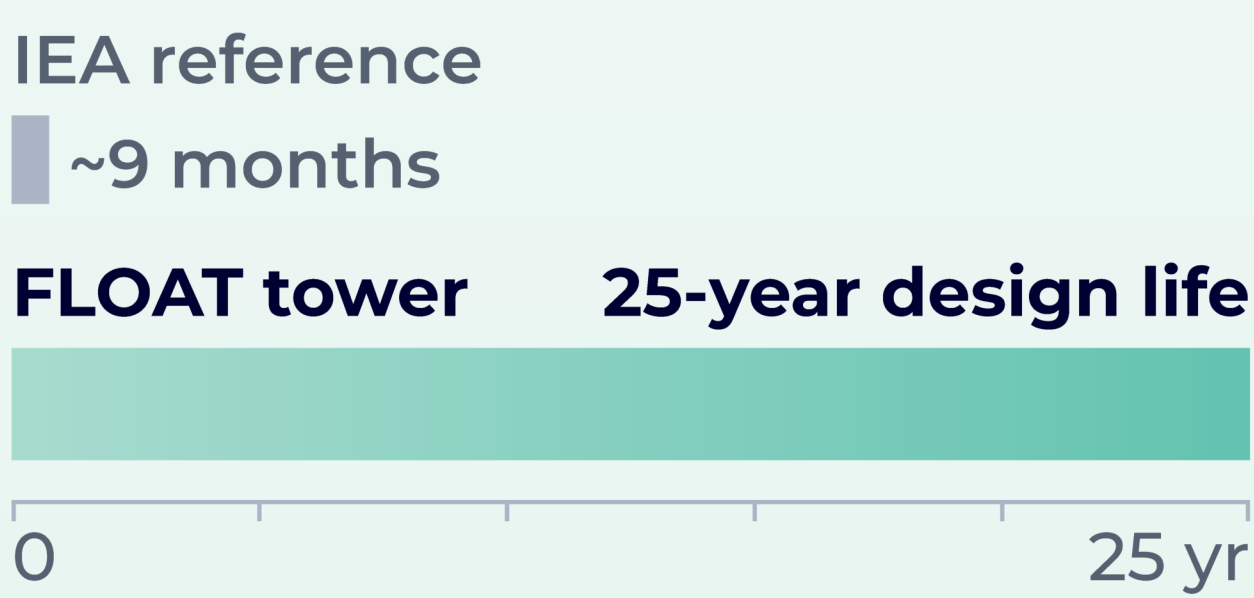
2 BENCHMARK FLOATBench [3]

Test whether AI surrogates hold outside their training data.

4 MONITOR FLOATSense [5]

Estimate tower loads at 11 heights from tower-top sensors.

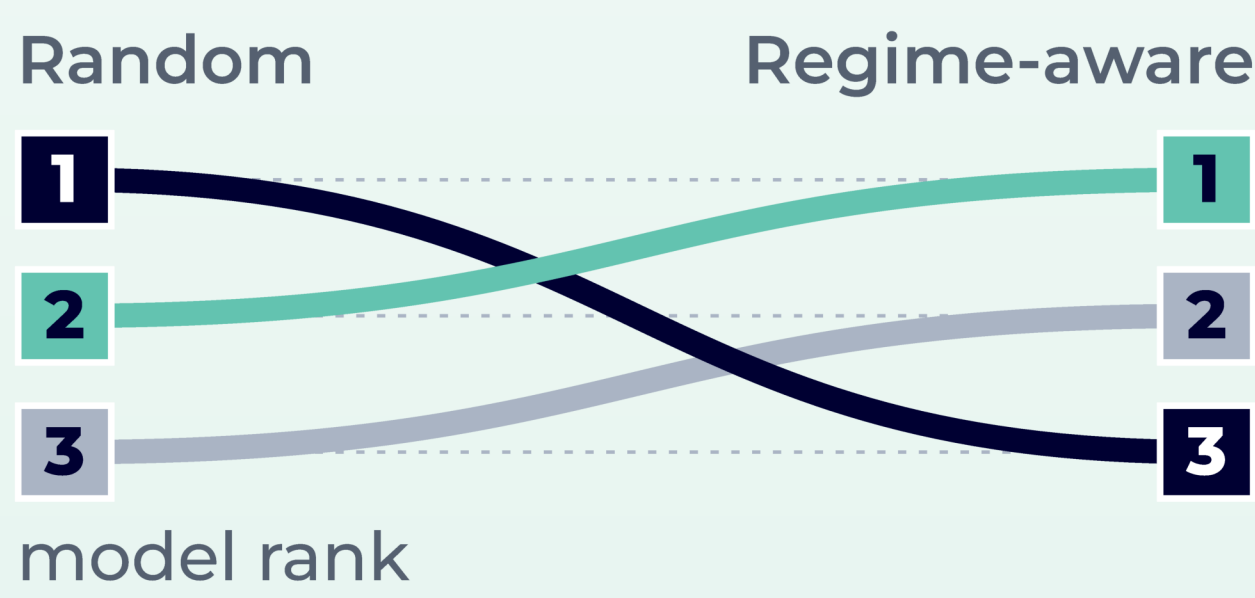
1 FLOAT [2]



25 years tower fatigue life (from ~9 months)

Merged into the official IEA 22 MW model (NREL/DTU).

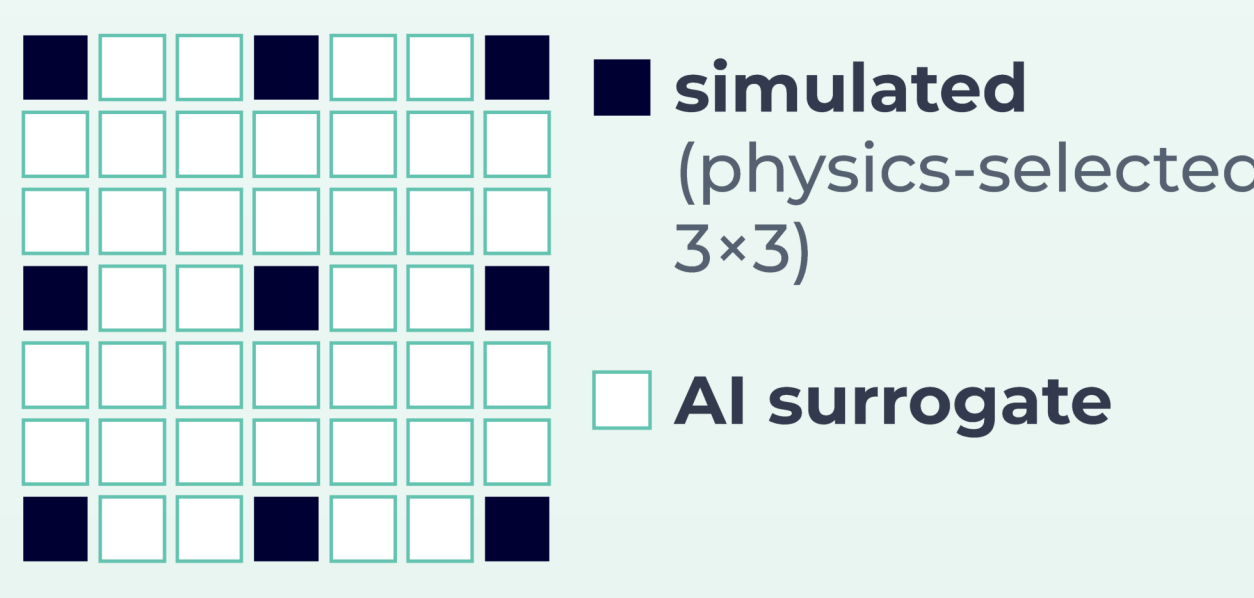
2 FLOATBench [3]



582,120 damage-labelled samples, 3 towers

Model rankings can invert when extrapolating.

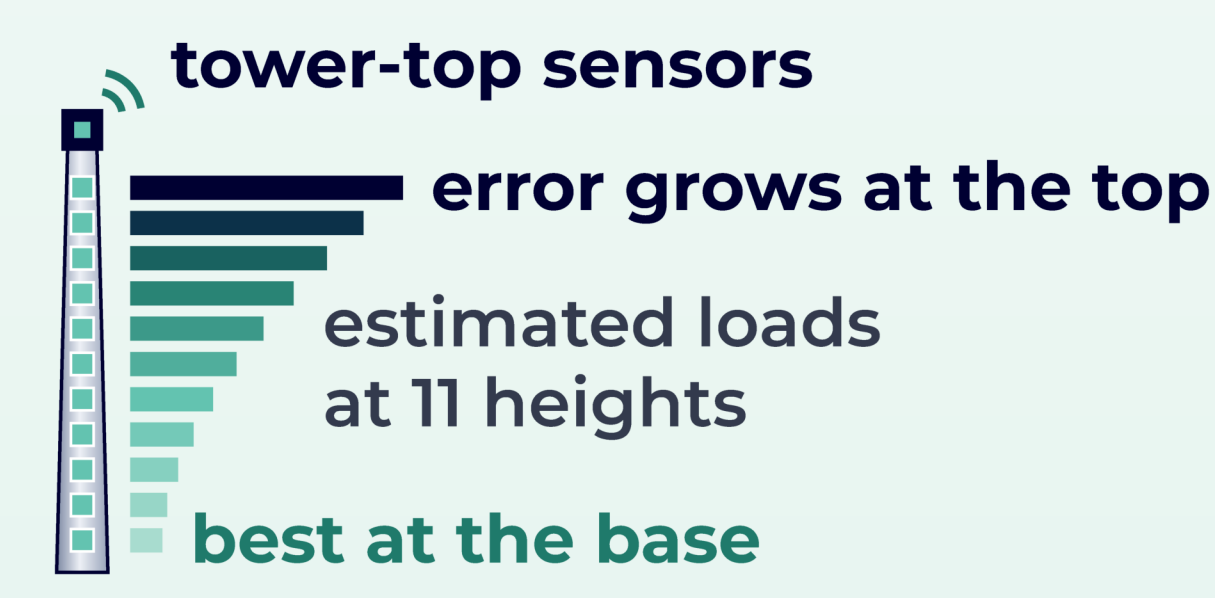
3 NeuroFLOAT [4]



5.4x fewer simulations (6,468 to 1,188)

Lifetime damage within ~0.5% of the full campaign.

4 FLOATSense [5]



11 heights loads from tower-top sensors

Best model at the base can lose accuracy at the top.

FOR INDUSTRY

- Before installation: far fewer simulations.
 - After installation: loads from tower-top sensors.
- (simulation-validated; field next)

PORTUGAL-US PARTNERSHIP

PORTUGAL
University of Porto (FEUP) | University of Aveiro
UNITED STATES
MIT | Brown University
INTERNATIONAL
TU Delft | IEA Wind

KEY MESSAGE

Physics provides reliability, AI provides scale.



ALL PROJECTS

References [1] Alves Ribeiro et al., Applied Energy 397 (2025) 126294. [2] FLOAT, Ocean Engineering 362 (2026) 126285. [3] FLOATBench, [4] NeuroFLOAT, [5] FLOATSense, all under review. MIT Portugal Seed Project: <https://doi.org/10.54499/2024.14532.MIT>.

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